

Customer Journey Map

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	2 Marks

Customer Journey Stages & Operational Flow:

The customer journey tracks a user's experience across 5 foundational stages: Onboarding, Activity Logging, Workout Search, AI Routine Recommendation, and AI Progress Review.

Stage of Journey	User Action	Thinking / Expectation	Touchpoints	Emotional State
Stage 1: Onboarding & Auth	Registers account with name, email, password; submits credentials to /api/auth/login.	'I want a fast, hassle-free registration and secure login credentials.'	POST /api/auth/register POST /api/auth/login	Neutral to Positive (Smooth access)
Stage 2: Workout Logging	Submits workout name, category, duration, and calories to POST /api/workouts.	'Will my workout be logged accurately with date and duration attributes?'	POST /api/workouts Mongoose Workout Model	Positive (Instant confirmation)
Stage 3: Workout Search & Query	Searches historical exercises by keyword (?q=running) or date to review performance.	'I want to quickly find my past running or cardio sessions without paging.'	GET /api/workouts/search GET /api/workouts	Positive (Fast regex query)
Stage 4: AI Workout Plan	Sends age, fitness goal, and experience level to request a personalized routine.	'I want an AI plan customized to my goal without paying for a coach.'	POST /api/ai/workout-recommendation Google Gemini 1.5 Flash	Delighted (Tailored guidance)
Stage 5: AI Fitness Insights	Submits total workouts, avg duration, and calories to evaluate progress.	'Is my consistency on track to hit my weight loss or strength milestones?'	POST /api/ai/fitness-insights Gemini Analytics Engine	Empowered (Actionable metrics)

Customer Journey Map (Continued) — Governance & Opportunities

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Process Ownership & Team Governance:

Process Ownership	Who is in the lead on this?	Core Responsibilities & Delivery KPI
Authentication & Security	Devanand R (Security Specialist)	Salted bcrypt password hashing, JWT token issuance, Bearer auth.js middleware guard.
Workout Engine & Search	Abishek M (Backend Lead)	Full workout CRUD controllers, Mongoose date sorting, multi-criteria regex search logic.
Generative AI Integration	Madhanraj R K (Team Leader & AI Architect)	Google Gemini SDK integration, fitness prompt engineering, workout plan & insights logic.
Database Architecture	Kishore samuvel S (Database Engineer)	MongoDB Atlas cluster configuration, User and Workout schemas, ObjectId referencing.
QA Automation & DevOps	Sunil kumar M (QA & DevOps Lead)	Comprehensive Postman test collection, centralized errorHandler.js, load latency benchmarking.

Journey Friction Points & Strategic Opportunities:

Journey Phase	Friction Point / Risk	Strategic Solution Implemented in AI FitTrack API
Onboarding	Weak passwords or duplicate email collisions.	Schema-level email uniqueness and bcrypt 10 salt rounds hashing.
Workout Logging	Invalid categories or missing required fields.	Mongoose schema enums for categories and strict field presence validation.
Workout Search	Complex multi-word search queries failing.	Case-insensitive MongoDB regex matching name, category, and date formats.
AI Routine	Slow LLM latency or hallucinated exercises.	Using gemini-1.5-flash model with concise, constraint-bound system prompts.
AI Insights	Misleading advice from empty metrics.	Pre-flight validation ensuring workout counts and duration are positive numbers.